MIL-R-45699(Ord)
10 May 1961
SUPERSEDING
FA(FC)-PD-532
4 September 1959

MILITARY SPECIFICATION

REFERENCE UNIT, HEIGHT: 8644098

1. SCOPE

1.1 This specification covers one type of height reference unit used in the Gun Direction Computers: M15 and M15C which are a part of the Field Artillery Fire Control Systems: M35 and M35C. The height reference unit is used to resolve a mechanical input of ORIGIN into two electrical outputs of REFERENCE and REPLOT (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitations for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1220

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

DRAWINGS

Ordnance Corps

D8643891	Electrical Schematic, Reference Units
F8644085	Refernece Units

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one height reference unit, representative of production, shall be made available to the Government to determine that the reference unit meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The height reference unit shall be manufactured in accordance with Drawing F8644085, "Reference Units", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3.1 Ground potential.- The design and construction of the height reference unit shall insure that all external parts, exclusive of electrical connections, are at ground potential at all times.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

3.5 Environmental.

3.5.1 Vibration.- The reference unit shall be subjected to a simple harmonic motion having an amplitude of 0.015 -inch (0.03 - inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second and return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the reference unit shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.5.2 Operating temperatures.- The height reference unit shall meet the requirements of 3.6 through 3.12 when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.5.3 Storage temperatures.- The height reference unit shall show no evidence of physical failure and shall meet the requirements of 3.6 through 3.12 at room temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 65 degrees Fahrenheit.

3.6 Input knob.- The input knob shall rotate in a clockwise direction for increasing values of northing.

3.6.1 Scale factor.- One revolution of each of the right digit counters shall indicate a change of 10 meters.

3.7 Torque.- The torque required for the rotation of the input knob through the entire range, as recorded by the counter readings, shall be between 10 and 20 ounce-inches at ambient room temperature (60 to 90 degrees Fahrenheit) and at the extreme temperatures specified in 3.5.2. Slippage of the knob shall occur between 2 and 3 pound-inches.

3.8 Backlash.- As a result of rotation of the input knob, backlash on the counter gear train shall be no greater than the smallest subdivision on the counter dial.

3.9 Shift-knob.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

3.9.1 ORIGIN position.- When the shift-knob is in the CCW (counterclockwise) detent position, indicating ORIGIN, the input knob shall engage the ORIGIN, REFERENCE and REPLOT counters. Rotation of the input-knob shall rotate all three counters simultaneously at the same rate. The reference and replot variable resistor shafts shall remain stationary. The ORIGIN and REFERENCE counters shall be illuminated. The REPLOT lamp shall be off.

3.9.2 REFERENCE position.- When the shifter knob is in the intermediate detent position indicating REFERENCE, the flap over the ORIGIN counter shall cover the counter and the input knob shall engage and operate only the REFERENCE counter. The input knob shall be capable of rotating through at least plus or minus 1,497 meters from the zero setting independent of the ORIGIN and REPLOT counters. While in this position the REFERENCE counter shall be illuminated and the REPLOT lamp shall be off.

3.9.3 REPLOT position.- When the shifter knob is in the CW (clockwise) detent position indicating the REPLOT position, the knob shall engage the REPLOT counter, and the ORIGIN and REFERENCE counters shall not function when the input knob is rotated to at least plus and minus 1,497 meters from the setting then appearing on the REFERENCE counter. While in this position the REPLOT counter shall be illuminated and the ORIGIN and REFERENCE lamps shall be off.

3.10 Electrical.

3.10.1 Circuitry.- Circuitry shall be in accordance with electrical schematic, Drawing D8643891.

3.10.2 Reference voltages.- The reference voltages applied to the height reference unit shall be 14.75 and 15.25 volts, 400 cps, phased instantaneously positive and negative as referred to a common neutral. The reference voltages shall be applied to the upper and lower limits of the reference and replot variable resistors through trimming variable resistors so that the reference voltages may be set at 15 volts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

3.11 Output accuracy.- With an applied input voltage, as depicted on the electrical schematic, the electrical output signal shall be linearly proportional to the mechanical angular rotation of the reference and replot variable resistors when loaded with 250 K ohms plus or minus 5 percent from wipers to the centertaps of the variable resistors. Under this condition the output voltages and counter readings in Table I shall be within the tolerances specified as a result of introducing the respective reference and replot counter readings by rotation of the input knob.

TABLE I

Reference and Replot Counter Readings (meters)	Output Voltages (volts)	Tolerances (millivolts)
1200 ± 2	12.00	± 15
800 ± 2	8.00	± 10
400 ± 2	4.00	± 10
0 ± 2	0.00	± 5
9600 ± 2	4.00	± 10
9200 ± 2	8.00	± 10
8800 ± 2	12.00	± 15

3.12 Illumination of counters.- The lamps for the windows of the counters shall be illuminated when energized from a plus 28-volt dc source and shall provide enough illumination to make all meter readings legible throughout the entire ranges. Illumination of the lamps shall be controlled by the shift-knob.

3.13 Alinement and calibration.- Alinement and calibration of all height reference units shall be performed using the same power source to insure uniformity of product. The reference voltages supplied shall be from a power source maintained as a standard.

3.14 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions.- The quality assurance provisions of referenced documents, to the extent specified herein, together with provisions of this section are applicable to inspection performed by the contractor and for Government verification.

4.1.1 The contractor shall provide an inspection system in accordance with Specification MIL-I-45208.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

4.2 Production model.- Unless otherwise specified in the contract, one height, reference unit shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance to drawing requirements prior to acceptance. These characteristics are defined as drawing requirements affecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the reference unit. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.4.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

TABLE II - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.4.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.5 Material.- Control for requirements of material shall be in accordance with Test Data requirements of Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.2 Accuracy requirements for special testing devices.

4.6.2.1 Device to test vibration requirements.- The device shall consist of a vibrating machine which will produce motion as specified. It shall be equipped with a medium capable of holding the unit securely to the vibrating machine in the specified positions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

4.6.2.2 Device to test torque requirements.- The device shall consist of a torque wrench type tester which will permit reading in both clockwise and counterclockwise directions with a minimum range of from zero to twenty-five ounce-inches, the accuracy of which shall be within plus or minus one-quarter pound-inch throughout the entire zone of excursion. The device shall be equipped with an adapter to accommodate the input knob. The device for testing slippage of the knob shall be similar but with a minimum range of zero to five pound-inches, the accuracy of which shall be within plus or minus one-quarter pound-inch throughout the entire zone of excursion. The device shall accommodate the same adapter used for the above knob test.

4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables III and IV shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%	Req't Par.	QA. Par.	Test Method
101. Post vibration	3.5.1	4.7.2	Device, Special Testing (DST)
102. Operating temperatures	3.5.2	4.7.3	DST

MINOR: NONE DEFINED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

TABLE IV- CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		Req't Par.	QA. Par.	Test Method
101.	Input knob	3.6	4.7.5	Visual
102.	Scale factor	3.6.1	4.7.6	Visual
103.	Torque	3.7	4.7.7	Device, Special Testing (DST)
104.	Backlash	3.8	4.7.8	Standard Measuring Equipment (SME)
105.	Origin position	3.9.1	4.7.9.1	Visual
106.	Reference position	3.9.2	4.7.9.2	Visual
107.	Replot position	3.9.3	4.7.9.3	Visual
108.	Circuitry	3.10.1	4.7.10	Visual
109.	Output accuracy	3.11	4.7.11	SME
110.	Illumination of counters	3.12	4.7.12	Visual
111.	Identification marking	3.14	4.7.13	Visual

MINOR: NONE DEFINED

4.7.2 Vibration.- This test shall be conducted in accordance with Method 201A of Standard MIL-STD-202, except that the time period for applied motion in each direction shall be as specified herein. Tests and performance measurements prior to or during vibration are not required for final acceptance. Subsequent to vibration, the reference unit shall be subjected to a visual-tactile examination for damage. The unit shall then be subjected to all the tests in this specification.

4.7.3 Operating temperatures.- This test shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" or raised beyond condition "G". When the reference unit is stabilized at each indicated temperature, it shall be subjected to the tests specified in 4.7.5 to 4.7.12 inclusive and shall meet the requirements specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

4.7.4 Storage temperature.- This test shall be conducted in accordance with Procedure II, condition "B" and "H" of Specification MIL-F-13926. Upon completion of the storage temperature cycling, the reference unit shall be subjected to tests specified in 4.7.5 to 4.7.12 inclusive, and shall meet the requirements specified. This test shall be performed on the production model only.

4.7.5 Input knob.- When the shifter knob is oriented to the ORIGIN, REFERENCE and REPLOT detent positions, respectively, clockwise rotations of the input knob shall cause the appropriate counters to record increasing values of height. A visual check shall be made to ascertain if this condition has been met.

4.7.6 Scale factor.- This test shall be performed with the shifter knob oriented to the ORIGIN, REFERENCE and REPLOT detent positions, respectively. With backlash excluded for direction of travel, rotate the input knob one complete revolution. This rotation of the input knob shall cause the appropriate counter to change the specified number of meters.

4.7.7 Torque.- This test shall be performed with the aid of special testing devices, affixed to the input knob. Orient the shifter knob to the ORIGIN detent position, then rotate the input knob through the entire range as indicated on the counters. The torque required for this rotation of the input knob and for clutch slippage shall be within the values specified. Repeat this test with the shifter knob oriented to the REFERENCE and the REPLOT positions.

4.7.8 Backlash.- Orient the shifter knob to the ORIGIN detent position. With backlash excluded for input knob rotation in a clockwise direction, record position of knob and counters. Rotate the input knob one complete revolution clockwise, then with backlash included, rotate the input knob one complete revolution counterclockwise without overtravel and record counter readings. The difference between the first and second counter readings is the amount of backlash in the gear train. Repeat this test with the shifter knob in the REFERENCE and the REPLOT detent positions. Backlash shall be read on the appropriate counter. Backlash in each counter gear train shall not exceed the tolerance specified.

4.7.9 Shifter knob.- The following tests shall be conducted with the lamp circuit energized from a 28 volt dc source.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

4.7.9.1 ORIGIN position.- Orient the shifter knob to the ORIGIN detent position. Visually ascertain that all digits of the ORIGIN and REFERENCE counters are sufficiently illuminated and that the REPLOT counter lamp is extinguished. Verify that a clockwise and counterclockwise rotation of the input knob shall rotate, without limits, the three counters as specified and that the reference and replot resistor shafts remain stationary during this operation. Return the counter to zero meters upon completion of this test.

4.7.9.2 REFERENCE position.- When the shifter knob is oriented to the REFERENCE detent position, the ORIGIN and REPLOT counters shall remain at their previous setting. Visually ascertain that the flap covers the ORIGIN counter, that all digits of the REFERENCE counter are sufficiently illuminated, and the REPLOT counter lamp is extinguished. Verify that a clockwise (plus) and a counterclockwise (minus) rotation of the input knob shall be capable of rotating the REFERENCE counter through the specified number of meters, and that the replot resistor shaft remains stationary.

4.7.9.3 REPLOT position.- When the shifter knob is oriented to the REPLOT detent position, the ORIGIN and REFERENCE counters shall remain at their previous setting. Visually ascertain if all digits of the REPLOT counter are sufficiently illuminated, and that the ORIGIN and REFERENCE counter lamps are extinguished. Verify that a clockwise (plus) and a counterclockwise (minus) rotation of the input knob shall be capable of rotating the REPLOT counter through the specified number of meters, and that the reference resistor shaft remains stationary.

4.7.10 Circuitry.- An electrical point-to-point check using SME shall be made to assure that ohmic and capacitive values conform to the prescribed electrical schematic.

4.7.11 Output accuracy.- Apply the specified reference voltages and resistance to the designated points of the variable resistors. Orient the shifter knob to the ORIGIN detent position. When the input knob is rotated to obtain the counter readings specified in Table I, the measured output voltages shall be within the tolerance specified. The output voltages shall be measured by use of standard measuring equipment.

4.7.12 Illumination of counters.- This test shall be performed in conjunction with the tests specified in 4.7.9. All digits of the counters shall be legible, when observed without the aid of magnification, in a darkened area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

4.7.13 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The reference unit shall be cleaned, preserved, packed, and marked in accordance with Specification MIL-F-14619/P8531. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The height reference unit is a functional plug-in subassembly of the gun direction computers (see 1.1) intended to receive a single, rotary, manual input of height by means of a mechanical gearing actuated by a rotary knob. The input is then converted to two electrical outputs: one designated REFERENCE, corresponding to the difference between the height component of origin coordinates and the height component of reference coordinates; the other signal designated REPLOT, corresponding to the differences between the height component of origin coordinates and the height component of the coordinates of a point being replotted.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable level of preservation, packaging and packing
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.

6.4.2 Special testing device (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.4.3 Tactile.- Skillful manipulation or sensing.

6.4.4 Production model.- A model in its final mechanical and electrical form of final production design made by production methods, tools, jigs and fixtures.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

Custodian:

Army-Ord

Preparing activity:

Army-Ordnance

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:47Z
Check the source to verify that this is the current version before use.